

Check-In 08/20. (True/False) The integral $\int (x-1)^2(x+5)^3 dx$ can be computed as a ‘shifting’ integral.

Solution. The statement is *false*. One wants to multiply the two terms in the integrand without explicitly expanding—but the non-matching powers prevents this. However, this might be possible with a ‘shift.’ There are only two possible choices for u to ‘shift’ the integrand: $u = x - 1$ (so that $x = u + 1$) or $u = x + 5$ (so that $x = u - 5$). In both cases, we have $du = dx$. In the first case, the resulting integral is $\int u^2(u+6)^3 du$, and in the second case, the resulting integral is $\int (u-6)^2 u^3 du$. In either case, we still cannot simply multiply the integrands without explicit expanding. To compute the given integral, there are several options. The most obvious is simple expansion:

$$\begin{aligned}\int (x-1)^2(x+5)^3 dx &= \int (x^5 + 13x^4 + 46x^3 - 10x^2 - 175x + 125) dx \\ &= \frac{x^6}{6} + \frac{13x^5}{5} + \frac{23x^4}{2} - \frac{10x^3}{3} - \frac{175x^2}{2} + 125x + C\end{aligned}$$

Alternatively, we can use integration-by-parts—particularly tabular integration (as each possible term in the integrand differentiates to 0 and can be repeatedly integration). The ‘harder’ choice of tabular integration yields...

u	dv
$(x+5)^3$	$(x-1)^2$
$3(x+5)^2$	$\frac{(x-1)^3}{3}$
$6(x+5)$	$\frac{(x-1)^4}{12}$
6	$\frac{(x-1)^5}{60}$
0	$\frac{(x-1)^6}{360}$

Therefore, we have...

$$\int (x-1)^2(x+5)^3 dx = \frac{(x+5)^3(x-1)^3}{3} - \frac{(x+5)^2(x-1)^4}{4} + \frac{(x+5)(x-1)^5}{10} - \frac{(x-1)^6}{60} + C$$

Alternatively, we have...

u	dv
$(x-1)^2$	$(x+5)^3$
$2(x-1)$	$\frac{(x+5)^4}{4}$
2	$\frac{(x+5)^5}{20}$
0	$\frac{(x+5)^6}{120}$

Therefore, we have...

$$\int (x-1)^2(x+5)^3 dx = \frac{(x-1)^2(x+5)^4}{4} - \frac{(x-1)(x+5)^5}{10} + \frac{(x+5)^6}{60} + C$$

Observe that none of these answers are equal to each other but instead differ by a constant.

Check-In 08/25. (True/False) To integrate $\int xe^{-3x^2} dx$, one uses LIATE to choose $u = x$ and $dv = e^{-3x^2}$.

Solution. The statement is *false*. To compute this integral with this choice of u and dv , one would have to integrate e^{-3x^2} . However, the function e^{-x^2} (and hence e^{-3x^2} , which differs only by a linear u -substitution of $x = \sqrt{3}u$, i.e. $u = x/\sqrt{3}$) has no elementary antiderivative, i.e. is not integrable. So, this approach will not work. However, we can integrate this with a simple u -substitution. Choosing $u = -3x^2$, we have $du = -6x dx$. But then...

$$\int xe^{-3x^2} dx = -\frac{1}{6} \int -6xe^{-3x^2} dx = -\frac{1}{6} \int e^u du = -\frac{e^u}{6} + C = -\frac{e^{-3x^2}}{6} + C$$

We can easily verify this as $\frac{d}{dx} \left(-\frac{e^{-3x^2}}{6} + C \right) = -\frac{e^{-3x^2}}{6} \cdot -6x + 0 = xe^{-3x^2}$.

Check-In 08/27. (True/False) The integral $\int \sin(2x) \cos(3x) dx$ can be treated as a 'looping integral.'

Solution. The statement is *true*. We can choose either $u = \sin(2x)$ or $u = \cos(3x)$. Choosing $u = \sin(2x)$, we have $dv = \cos(3x)$. We then have...

u	dv
$\sin(2x)$	$\cos(3x)$
$2 \cos(2x)$	$\frac{\sin(3x)}{3}$
$-4 \sin(2x)$	$-\frac{\cos(3x)}{9}$

So that...

$$\int \sin(2x) \cos(3x) dx = \frac{\sin(2x) \sin(3x)}{3} + \frac{2 \cos(2x) \cos(3x)}{9} + \frac{4}{9} \int \sin(2x) \cos(3x) dx$$

$$\frac{5}{9} \int \sin(2x) \cos(3x) dx = \frac{\sin(2x) \sin(3x)}{3} + \frac{2 \cos(2x) \cos(3x)}{9}$$

$$\int \sin(2x) \cos(3x) dx = \frac{9}{5} \left(\frac{\sin(2x) \sin(3x)}{3} + \frac{2 \cos(2x) \cos(3x)}{9} \right) + C$$

$$\int \sin(2x) \cos(3x) dx = \frac{1}{5} (3 \sin(2x) \sin(3x) + 2 \cos(2x) \cos(3x)) + C$$

Alternatively, choosing $u = \cos(3x)$ and $dv = \sin(2x)$, we have...

u	dv
$\cos(3x)$	$\sin(2x)$
$-3 \sin(3x)$	$-\frac{\cos(2x)}{2}$
$-9 \cos(3x)$	$-\frac{\sin(2x)}{4}$

So that...

$$\int \sin(2x) \cos(3x) dx = -\frac{\cos(3x) \cos(2x)}{2} - \frac{3 \sin(3x) \sin(2x)}{4} + \frac{9}{4} \int \cos(3x) \sin(2x) dx$$

$$-\frac{5}{4} \int \sin(2x) \cos(3x) dx = -\frac{\cos(3x) \cos(2x)}{2} - \frac{3 \sin(3x) \sin(2x)}{4}$$

$$\int \sin(2x) \cos(3x) dx = -\frac{4}{5} \left(-\frac{\cos(3x) \cos(2x)}{2} - \frac{3 \sin(3x) \sin(2x)}{4} \right) + C$$

$$\int \sin(2x) \cos(3x) dx = \frac{1}{5} (2 \cos(3x) \cos(2x) + 3 \sin(3x) \sin(2x)) + C$$

Alternatively, one can use the product-to-sum formula:

$$\sin A \cos B = \frac{\sin(A+B) + \sin(A-B)}{2}$$

We then have...

$$\begin{aligned} \int \sin(2x) \cos(3x) dx &= \int \frac{\sin(2x+3x) + \sin(2x-3x)}{2} dx \\ &= \frac{1}{2} \int (\sin(5x) + \sin(-x)) dx \\ &= \frac{1}{2} \left(-\frac{\cos(5x)}{5} + \cos(-x) \right) + C \\ &= \frac{1}{2} \left(\cos x - \frac{\cos(5x)}{5} \right) + C \\ &= \frac{1}{10} (5 \cos x - \cos(5x)) + C \end{aligned}$$

Check-In 09/01. (True/False) To integrate $\int \tan^7 \theta \sec^5 \theta d\theta$, one should choose $u = \sec \theta$.

Solution. The statement is *true*. Observe that the ‘only’ choices are $u = \tan \theta$ or $u = \sec \theta$. If one chooses $u = \tan \theta$, we have $du = \sec^2 \theta d\theta$. But then there would remain three terms of $\sec \theta$ in the integrand, and the Pythagorean Identities only allow one to replace even powers of $\sec \theta$. However, choosing $u = \sec \theta$, we have $du = \sec \theta \tan \theta d\theta$. From the Pythagorean Identities, we have $\tan^2 \theta + 1 = \sec^2 \theta$, i.e. $\tan^2 \theta = \sec^2 \theta - 1$. But then $\tan^6 \theta = (\tan^2 \theta)^3 = (\sec^2 \theta - 1)^3$. Therefore, we have...

$$\int \tan^7 \theta \sec^5 \theta d\theta = \int \tan^6 \theta \sec^4 \theta \cdot \sec \theta \tan \theta d\theta = \int (\sec^2 \theta - 1)^3 \sec^4 \theta \cdot \sec \theta \tan \theta d\theta = \int (u^2 - 1)^3 u^4 du$$

But then, we have...

$$\int (u^2 - 1)^3 u^4 du = \int (u^{10} - 3u^8 + 3u^6 - u^4) du = \frac{u^{11}}{11} - \frac{u^9}{3} + \frac{3u^7}{7} - \frac{u^5}{5} + C$$

Therefore, we have...

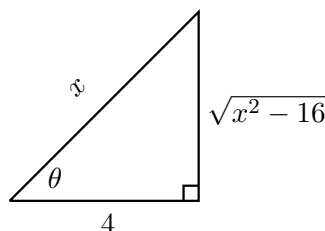
$$\int \tan^7 \theta \sec^5 \theta d\theta = \frac{\sec^{11} \theta}{11} - \frac{\sec^9 \theta}{3} + \frac{3 \sec^7 \theta}{7} - \frac{\sec^5 \theta}{5} + C$$

Check-In 09/03-1. (True/False) One can integrate $\int \frac{7x}{(x^2 - 16)^5} dx$ using trig. substitution; however, this would not be an ‘optimal’ approach.

Solution. The statement is *true*. There is commonly overlap between different techniques of integration; however, this does not mean that the methods are equally ‘simple.’ For instance, the given integral can be integrated using u -substitution: let $u = x^2 - 16$, so that $du = 2x dx$. But then...

$$\int \frac{7x}{(x^2 - 16)^5} dx = \frac{7}{2} \int \frac{2x}{(x^2 - 16)^5} dx = \frac{7}{2} \int \frac{du}{u^5} = \frac{7}{2} u^{-4} \cdot \frac{-1}{4} + C = -\frac{7}{8(x^2 - 16)^4} + C$$

Observe that this can also be integrated using integration-by-parts. We choose $u = 7x$ and $dv = \frac{1}{(x^2 - 16)^5}$, though the integration of dv requires partial fractions. One can also use partial fractions for the original integral. However, each of these is rather tedious and we will not produce these solutions here. Finally, one can use trig substitution.



We have $\sec \theta = \frac{x}{4}$, so that $x = 4 \sec \theta$. But then $dx = 4 \sec \theta \tan \theta d\theta$. We know $\tan \theta = \frac{\sqrt{x^2 - 16}}{4}$, so that $\sqrt{x^2 - 16} = 4 \tan \theta$. But then $(x^2 - 16)^5 = (x^2 - 16)^{10/2} = (\sqrt{x^2 - 16})^{10} = (4 \tan \theta)^{10}$.

Therefore, we have...

$$\int \frac{7x}{(x^2 - 16)^5} dx = \int \frac{7 \cdot 4 \sec \theta}{4^{10} \tan^{10} \theta} \cdot 4 \sec \theta \tan \theta d\theta = \frac{7}{4^8} \int \frac{\sec^2 \theta}{\tan^9 \theta} d\theta$$

We stop here because it is already evident that this is not ‘optimal’ in comparison to the u -substitution approach shown—which was the point.

Check-In 09/03-2. (True/False) The partial fraction decomposition of $\frac{x+3}{x(x^2+4)}$ has the form

$$\frac{A}{x} + \frac{B}{x^2+4}.$$

Solution. The statement is *false*. Recall that if the ‘base’ irreducible term is quadratic, then the corresponding factor in the numerator of the partial fraction decomposition is a linear term. Therefore, the correct partial fraction decomposition of this rational function is...

$$\frac{x+3}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$